Naomi R. Shvedov, PhD

Postdoctoral Associate · Sorrells Lab · University of Pittsburgh

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APPOINTMENTS

2025 – present **Postdoctoral Associate**, Sorrells Lab, Department of Neuroscience, University of Pittsburgh

EDUCATION

2019 – 2025 **Ph.D., Neuroscience**, Boston University
Dissertation: Dynamics and mechanisms of neuron migration in the adult songbird brain.
Advisor: Benjamin B. Scott.

2015 – 2019 **B.S., Cell Biology and Neuroscience**, Rutgers University
Graduated Magna cum laude. Rutgers Cell Biology and Neuroscience Honors Program.

HONORS, AWARDS, AND FELLOWSHIPS

2026 Center for the Neural Basis of Cognition (CNBC) Professional Development Award

2022 Chan Zuckerberg Initiative (CZI) Bioimaging North America Program Professional Development Award

2020 – 2022 NSF NRT Neurophotonics Fellowship, Boston University

PUBLICATIONS

Peer-reviewed:

Shvedov, N.R., Castonguay, S.J., Rother, A., Schick, D., Kornfeld, J., Scott, B. (2026). Songbird connectome reveals tunneling of migratory neurons in adult striatum. *Current Biology*. https://doi.org/10.1016/j.cub.2026.03.057

Shvedov, N.R., Analoui, S., Dafalias, T., Bedell, B.L., Gardner, T.J., Scott, B.B. (2024). In vivo imaging in transgenic songbirds reveals superdiffusive neuron migration in the adult brain. *Cell Reports*. vol. 43, issue 2. https://doi.org/10.1101/2023.07.14.548876

Mamounis, K.J., **Shvedov, N.R.**, Margolies, N., Yasrebi, A., Roepke, T.A. (2019). The effects of dietary fatty acids in the physiological outcomes of maternal high-fat diet on offspring energy homeostasis in mice. *Journal of Developmental Origins of Health and Disease*. pp. 1–12. doi:10.1017/s2040174419000540

Manuscripts in preparation:

Frostig, H., **Shvedov, N.R.**, Scott, B.B. (2025). Deep brain three-photon imaging in songbirds. *In preparation*.

Spotlights:

Shvedov, N.R., Sorrells, S.F. (2025). Why reinvent the wheel? An idea to watch: extending growth cones to migrating neurons. *BioEssays*. https://doi.org/10.1002/bies.70059

Shvedov, N.R., Margolis, D.J. (2019). An optical exposé of cortical function. *Trends in Neurosciences*. vol. 42, no. 8, pp. 511–513. doi:10.1016/j.tins.2019.05.003

INVITED TALKS

Jun 2024	CoNNExINs Symposium, New York University
Feb 2024	Neurophotonics Trainee Tech Talk, Boston University
May 2023	Neurodevelopment for Experiment–Theory Interaction Workshop, Harvard University
Mar 2023	Cosyne Workshop on Neural Development and Evolution, Mont Tremblant, QC
Aug 2022	Gordon Research Conference on Neural Development, Salve Regina University

CONFERENCE PRESENTATIONS

Apr 2026	**Shvedov, N.R.**, Bandi, A.C., Runyan, C.A., Sorrells, S.F. *Uncovering the role of a "late bloomer" — the paralaminar amygdala nucleus.* University of Pittsburgh Postdoctoral Association Symposium.
Oct 2024	**Shvedov, N.R.**, Castonguay, S.J., Rother, A., Schick, D., Kornfeld, J., Scott, B. *EM connectomics reveal tunneling of migratory neurons in adult songbird brain.* Janelia Conference for Developmental Specification of Complex Behaviors.
Nov 2023	**Shvedov, N.R.**, Frostig, H., Mertz, J., Scott, B.B. *Deep brain three-photon imaging in transgenic songbirds.* Society for Neuroscience Annual Meeting.
2022	**Shvedov, N.R.**, Gardner, T.J., Scott, B.B. *In vivo imaging in transgenic songbirds reveals superdiffusive neuron migration in the adult brain.* Presented at: Gordon Research Conference on Neural Development; Gordon Research Conference on Neural Mechanisms of Acoustic Communication; Society for Neuroscience Annual Meeting; UC Irvine CNCM Conference on the Structure, Function and Development of Neural Circuits.

ADDITIONAL TRAINING AND WORKSHOPS

Dec 2022	1st International Mini2P Workshop, Moser Group, Kavli Institute, Trondheim
Jun 2022	Frontiers in Neurophotonics Summer School, Université Laval

TEACHING AND MENTORING

2024 – 2025	**Teaching Fellow**, Introduction to MATLAB (NE/PS 212), Boston University. Led discussion sections, review sessions, and hands-on labs; taught four in-class lectures.
Fall 2023	**Teaching Fellow**, Principles of Neuroscience (GMS NE 700), Boston University. Supported first-year PhD students.
Summer 2022	**GROW Mentor**, Greater Boston Area Research Opportunities for Young Women. Mentored a high-school student in a STEM research project as part of a program supporting underrepresented students.

SERVICE AND OUTREACH

Feb 2026	**Neuroscience Proseminar Lecturer**, University of Pittsburgh.
Nov 2025	**Abstract Reviewer**, Cosyne 2026.
Jan – Feb 2024	**Graduate Student Member**, Biology Faculty Search Committee (Neuroethology), Boston University.

2020 – 2022	**Member**, NSF NRT Neurophotonics Training Program Seminar Planning Committee, Boston University. Coordinated faculty seminars on state-of-the-art neurophotonics.
2020 – 2021	**Recruitment Co-Chair**, Neuroscience Graduate Student Organization, Boston University. Led virtual recruitment for the Graduate Program of Neuroscience.
Oct 2020	**Mentor Judge & GPN Exhibitor**, SACNAS National Diversity in STEM Conference.

TECHNICAL SKILLS

In vivo	Chronic two- and three-photon calcium imaging through cranial windows and GRIN lenses in live mice; head-fixed behavior; stereotaxic and neonatal viral injection; chronic GRIN lens implantation in deep brain (amygdala); chronic chemogenetic manipulation.
Tissue & histology	Ex vivo organotypic slice culture (mouse and human); immunohistochemistry; AAV transduction, confocal and epifluorescence imaging.
Molecular	RNA extraction, reverse transcription, qPCR.
Analysis & computation	MATLAB; Python (NumPy/Pandas/Matplotlib); custom analytical pipelines for calcium imaging and cell tracking.
Engineering	3D modeling and printing of custom behavioral and head-fixation hardware.